

✓ SLIP – 13

Section I: Data Structures [15 Marks]

Q 1) Accept n sorted values in array from user. Accept a value x from user and use binary search algorithm to check whether the number is present in sorted array or not and output the position if the number is present.

Answer:

```
#include <stdio.h>

int main()
{
    int a[100], n, i, x, low, high, mid, pos = -1;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    printf("Enter %d sorted values:\n", n);
    for(i = 0; i < n; i++)        scanf("%d",
    &a[i]);

    printf("Enter value x to search: ");
    scanf("%d", &x);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
        mid = (low + high)
/ 2;

        if(a[mid] == x)
        {
            pos =
mid + 1;        break;
        }

        else if(a[mid] < x)
        low = mid + 1;        else
        high = mid - 1;
    }
}
```

```

    }

    if(pos != -1)        printf("Number found at
position %d", pos);    else        printf("Number
not found");

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Bus (Busno , capacity ,depot_name)

Route (Routeno ,source ,destination ,no_of_stations)

Relationship: Bus-Route : Many to one

Constraint: Bus capacity should not be null.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Route(
Routeno INT PRIMARY KEY,
source VARCHAR(50),
destination VARCHAR(50),
no_of_stations INT
);

CREATE TABLE Bus(      Busno
INT PRIMARY KEY,      capacity
INT NOT NULL,      depot_name
VARCHAR(50),
      Routeno INT REFERENCES Route(Routeno)
);

INSERT INTO Route VALUES
(41,'Pune station','Baramati',12),
(42,'Pimpri','Hadapsar',2),
(43,'Pimpri','Vimannagar',6);

```

```
INSERT INTO Bus VALUES
(201,40,'Swargate',41),
(204,15,'Sangamner',42),
(205,30,'Sangamner',43);
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. List all buses at depot "Swargate".

Answer:

```
SELECT * FROM Bus
WHERE depot_name = 'Swargate';
```

ii. Delete Bus whose Bus number is "204".

Answer:

```
DELETE FROM Bus
WHERE Busno = 204;
```

iii. List all buses on route number 41.

Answer:

```
SELECT * FROM Bus
WHERE Routeno = 41;
```

iv. List the route details having number of stations greater than 10.

Q 2(C)

C. Write a query to delete all the buses on routes in which total stations are less than 3. [4 marks]

Answer:

```
DELETE FROM Bus
WHERE Routeno IN
(
    SELECT Routeno FROM Route
    WHERE no_of_stations < 3
);
```

Q 3) Viva [5 marks]